



KU LEUVEN DIFFERENCE-IN-DIFFERENCES WORKSHOP

Staggered Adoption in Difference-in-Differences

A Callaway & Sant'Anna Checklist, Live, on a Real Project

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KU Leuven · Difference-in-Differences Workshop

This morning: an application, not a theorem

We will work through a single empirical project to illustrate the practical side of Callaway & Sant'Anna:

- (a) The recommended application of **CS diagnostics**.
- (b) **Software syntax problems** when estimating pre-trend coefficients.
- (c) **Covariate imbalance**, spotting it, and what to do about it.
- (d) **Checklists**, a disciplined sequence before you ever look at the outcome.
- (e) **Perfect separation** and propensity scores.

The vehicle is a project of mine with Christine Durrance and Melanie Guldi on online dating and its effect on birth rates.

A word on the data before we start

- The project has **changed significantly** since I built these slides.
- But everything you are about to see is **true**, a real dataset and a real treatment assignment.
- **None of this is simulated data.** The rollout, the imbalance, the event studies, all of it is the genuine empirical object.

LESSON

The Diff-in-Diff Checklist

How checklists saved lives in medicine

Key idea: checklists reduce errors and improve patient safety by standardizing procedures and preventing overlooked steps.

- **The Checklist Manifesto** (Atul Gawande) explores how checklists improve outcomes in medicine, aviation, and other fields.
- The **WHO Surgical Safety Checklist** reduced complications and mortality in surgeries worldwide, one study found a **36% reduction** in major surgical complications.
- Checklists are thought to work because they:
 - Ensure that critical steps are not skipped.
 - Encourage teamwork and communication.
 - Create a structured, repeatable process for complex tasks.

More discretion makes checklists more essential

- In yesterday afternoon's talk, "**AI Agents and the Minimum Wage**," I argued that the more discretion we give AI agents in research, the more they explore that space **autonomously**.
- That makes it *even more* critical that we use **checklists** when designing our research, in general, too.
- So today I'll use a checklist and an application to walk us through **staggered adoption** in difference-in-differences.
- I'll introduce the estimators **in the context of walking through a real project**, the effect of online dating on birth rates in the United States.

The 10-point checklist

1. **Define your target parameter**, potential outcomes, population, and weighted averages. What exactly are you trying to estimate, and who is your population? What is your research question asking, and how can it be expressed as a proper average treatment effect? Can you justify that decision to someone else?
2. **Create a table counting units by cohort and “treatment shares.”** Count the units treated in each cohort, including the “never treated” and “always treated” cohorts. Calculate the share of all treated units treated in each cohort.
3. **Plot treatment rollout.** Visualize when and how treatment varies across units and time. Consider `PanelView` by Yiqing Xu.

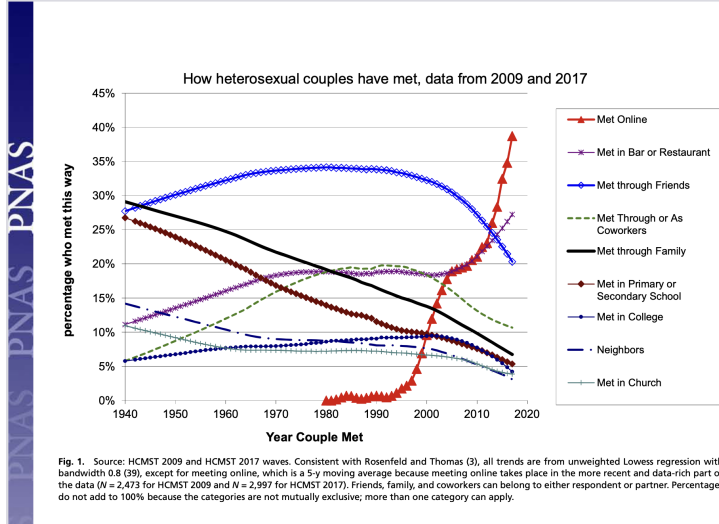
The 10-point checklist

4. **Pick your control group.** Decide which units serve as your counterfactual, and justify the choice. Not-yet-treated? Why? Never-treated? Why? Something else? Why?
5. **Choose between unconditional and conditional parallel trends.** Assess whether treated and control units would have followed similar paths absent treatment.
6. **Check covariate imbalance.** Examine whether observable characteristics differ systematically between treated and control groups. Check this a variety of different ways.

The 10-point checklist

7. **Plot average outcomes across cohorts.** Look at outcome trends by treatment timing to spot potential problems.
8. **Estimator selection and assumptions.** Choose the difference-in-differences estimator that fits your design, and understand what it assumes. On what basis: *covariates*, overlap vs. extrapolation.
9. **Check for parallel-trends violations** with falsifications, event studies, and sensitivity analysis. Test your identifying assumptions rigorously. Make sure you understand how your pre-trends are calculated, and confirm they are what you intended to calculate.
10. **DDDiD, or “Don’t Do Diff-in-Diff.”** Diff-in-diff needs parallel trends; if you lose confidence in that, consider moving to a different design, even synthetic control.

How heterosexual couples meet



Conceptual framework

- Online dating is thought to **reduce search frictions** and thicken romance markets.
 - It could create long-term partnerships where families are an intentional choice.
 - It could therefore **increase** birth rates for the marginal couple.
- But complaints have risen that online dating causes “**permanent dating**,” which could reduce births.
 - These are profit-maximizing, publicly traded firms with arguably concentrated ownership, an **HHI around 4,500** (Match owns ~75 of the platforms).
 - They may, purposefully or inadvertently, design interfaces to **minimize exit** from the market through romantic matches, exactly the “permanent dating” channel.

This is an old version of the study, but a real dataset with a real treatment assignment (not simulated). It is presented as a vehicle for thinking about staggered adoption, CS, and how diagnostics uncovered problems.

Checklist step 1: define the target parameter

Recall that all average treatment effects have **three ingredients**:

1. Potential outcomes representing treatment effects.
2. Populations of units at a point in time.
3. Weighted expectations.

So we have to make a **design** decision: which treatment effect will we target, and why?

Checklist step 1: define the target parameter

$$\text{Treatment effect: } \delta_i = Y_i(1) - Y_i(0)$$

- I start with a binary indicator and will explain momentarily which intervention I use.
- Don't think in terms of "the treatment effect," because there is no single "the."
- Imagine you had trillions of euros and permission to run any randomized trial you wanted.
- You would randomize **two** things, not one: $Y(1)$ is "received online dating."
- But what is $Y(0)$?
- In a trial the comparison group isn't given "no medicine," but a **different** treatment, usually a sugar pill.

Checklist step 1: define the target parameter

- You have to decide up front: what state of the world is the most relevant comparison for you?
- Let's look at the history of online dating to motivate the **different** parameters we could estimate.
- Consider two eras: the early **open-website** period and the late **swiping** period.

History of online dating

Timeline of Online Dating Platforms and their Purpose

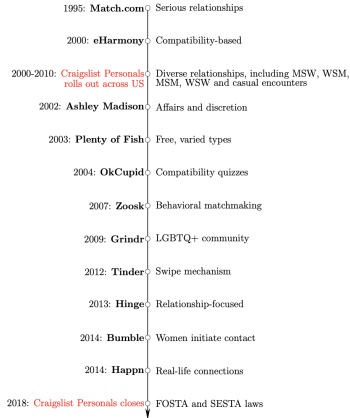


Figure 1: History of Major Online Dating Platforms

Checklist step 1: the open-website era

- In the 1990s and early 2000s, online dating was dominated by **Match, OKCupid, and eHarmony**.
- These were “open websites”: anyone could use them, so selection into use was deeply endogenous without external variation.
- You would need an instrument (a LATE), matching on observables, or something more parametric.
- The upside is that the counterfactual is more plausibly $Y(0) =$ a world **without** online dating, simply because it is so early.
- But in this era online dating was mainly a path to life partners, not casual dating.
- It was also less common, and often **stigmatizing** as a signal of “failure on the marriage market.”

Checklist step 1: the swiping era

- The later “swiping” era is different, and in many ways it is **the** contemporary platform people experience now.
- This is hard, because $Y(0)$ is probably **not** “a world without online dating”: that world has stopped existing.
- Much more plausibly, $Y(0)$ means “a **different** online dating platform.”
- This is also a period when online dating is much more common.
- After Tinder, this is the rise of **casual-sex** online dating.

Checklist step 1: picking your treatment effect

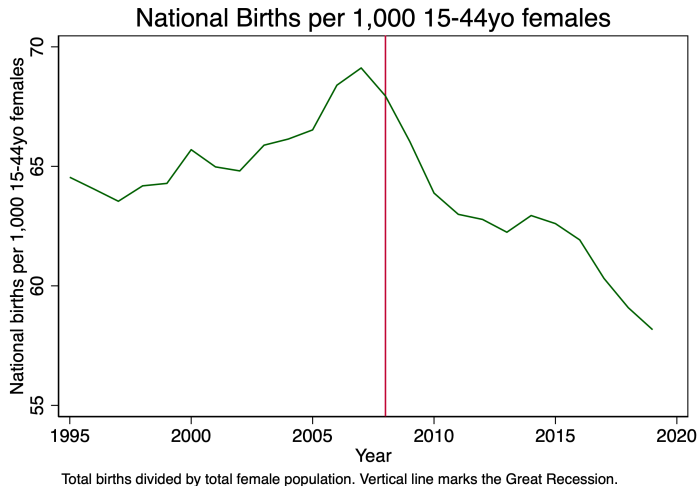
$$\delta = Y(\text{online dating}) - Y(\text{friends and family})$$

versus

$$\delta = Y(\text{online dating platform 1}) - Y(\text{online dating platform 2})$$

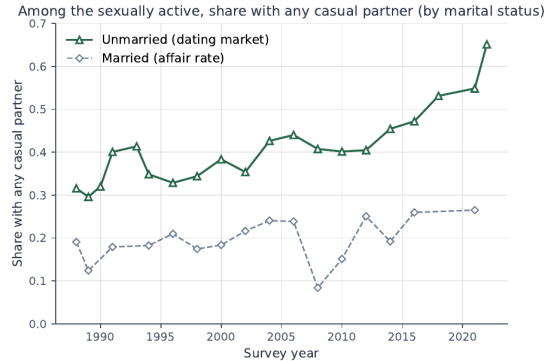
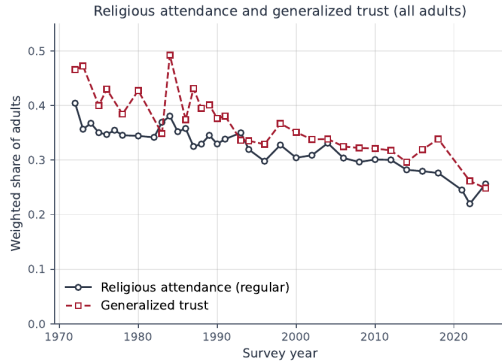
- You have to decide which of these you care about, and make your design decisions around it.
- I target the **first**, with $Y(\text{friends and family}) = 0$ as my counterfactual, which means going further back in time.
- I also go back because the post-2008 period is rife with **confounders**.

The national birth rate



The dating market has changed structurally since 1972

GSS long-arc indicators of dating-market structural change, 1972--2024

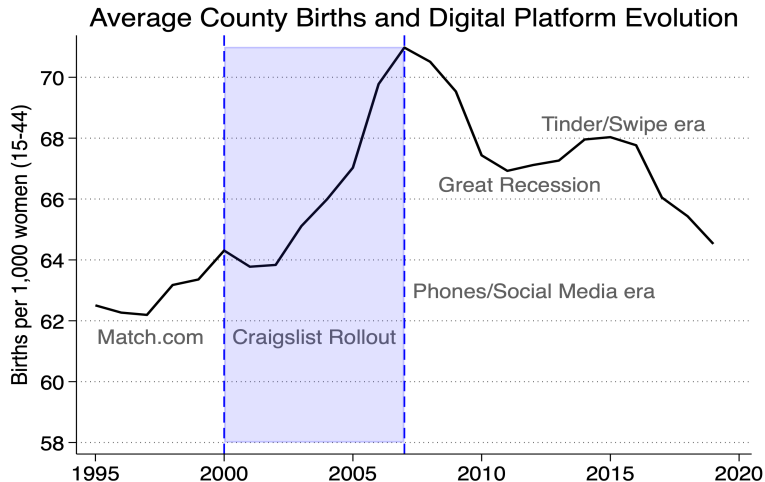


Confounders

Two problems:

1. Online dating either hits the US with an **open website** anyone can get on (early period) or a **“swiping app”** anyone can get on (late period).
2. After 2008, American birth rates **plummeted and never recovered** (demographic transition).
 - Both are massive, hurricane-like winds, and it's going to be challenging to deal with them with diff-in-diff.
 - But, amazingly, we have a solution that gets at *both* and is probably much closer to our target parameter, **Craigslist Personals**.

Birth rates and the digital dating timeline



Data selection

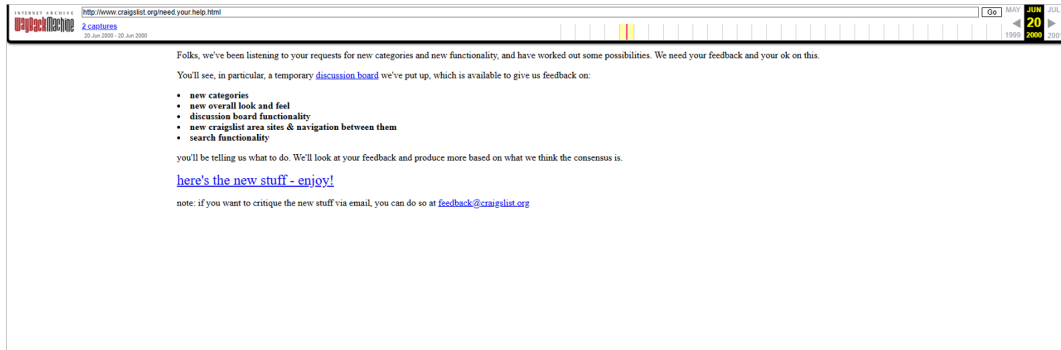
- We focused originally on **birth rates for 15–44yo females**; the unit of observation is the **county**.
- Sample period: **1995** (when Craigslist first opens) to **2007**. Why 2007? Three reasons:
 1. Birth rates plummeted following the Great Recession (demographic transition).
 2. Online dating apps like Tinder appear simultaneously, no geographic variation to exploit.
 3. We wanted to avoid the social-media environment.
- We keep counties treated in 2008–2010 in the dataset as a sample of **“not-yet-treated”** counties.
- Used the **Wayback Machine** to find the exact month Craigslist Personals appeared in each county, assigned it to adjacent counties too, then coded a county-year as treated if it was ever treated.

The Wayback Machine

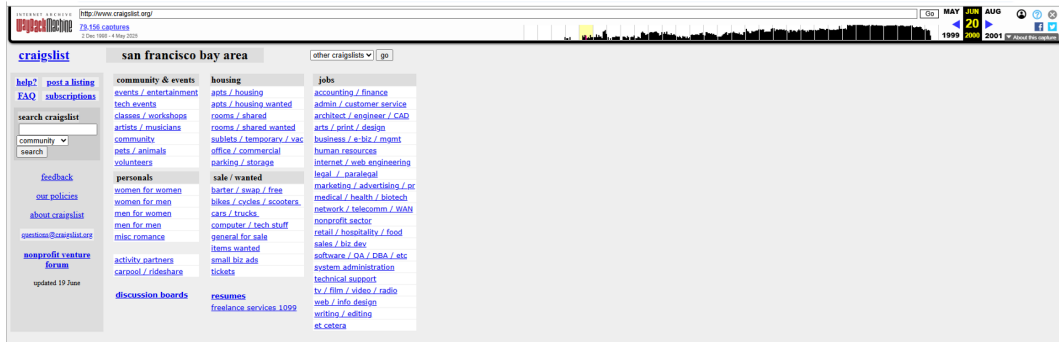
- In my JHR with Greg DeAngelo and John Tripp, we used the **Wayback Machine** to identify the start dates of Craigslist opening erotic services.
- Here we use it to identify the dates when **Craigslist Personals** showed up.
- We went through every US city to find the exact date it got Craigslist Personals.
- Personals potentially linked the old website era of romance-oriented online dating with the modern swipe era and its stronger **casual-matching** component.

Let's look at the history of Craigslist Personals.

The history of Craigslist Personals



The history of Craigslist Personals



The history of Craigslist Personals

INTEREST ARCHIVE
Wayback Machine
http://www.craigslist.org/
79,156 captures
2 Dec 1999 - 4 May 2025

Go JUL AUG OCT
15
1999 2000 2001

craigslist **san francisco bay area** other craigslists go

contact **post a listing**
FAQ **subscriptions**

search craigslist
community
search

discussion forums
antispam measures
our policies
about craigslist
craigslist nonprofit
venture forum
6/S party pix
updated 11 August

community
activity partners
artists / musicians
carpool / rideshare
childcare / kids
general community
pets / animals
volunteers

housing
acts / housing
acts / housing wanted
rooms / shared
rooms / share wanted
sublets / temporary
sublets wanted **new**
office / commercial
parking / storage

events
classes / workshops
events / entertainment
tech events

sale / wanted
barter / swap / free
bikes / cycles
cars / trucks
computer / tech stuff
general for sale
general wanted
women for men
men for women
men for men
misc romance

jobs
accounting / finance
admin / customer service
architect / engineer / cad
arts / print / design
business / e-biz / mgmt
education / teaching **new**
human resources
internet / web engineer/g
legal / paralegal
market's / advertising / pr
medical / health / biotech
network / telecom
nonprofit sector
retail / hospitality / food
sales / biz dev
skilled trade / craft **new**
software / QA / DBA
system administration
technical support
tv / film / video / radio
web / info design
writing / editing
et cetera

san francisco
south bay
east bay
north bay
peninsula
boston
new york city - **new**
DC - **new**
portland **ore** - **new**
seattle - **new**
los angeles - **new**
san diego - **new**
chicago - **new**
sydney
melbourne

The history of Craigslist Personals

INTERNET ARCHIVE Wayback Machine <http://www.craigslist.org/> 79.156.caatures 2 Dec 1998 - 4 May 2001

SEP OCT DEC 1999 2000 2001

craigslist **san francisco bay area** other craigslists go

contact **post a listing**
FAQ subscriptions

search craigslist
community search

discussion forums
flag for review new

our nonprofit work
spotlight: Precita Eyes

craigslist still seeking
new office space

addressing pranks,
spam & abuse

our policies
about craigslist
updated 12 october

community
activity partners
artists / musicians
carpool / rideshare
childcare / kids
general community
pets / animals
political forum new
volunteers

events
classes / workshops
events / entertainment
tech events

personals
women seeking women
women seeking men
men seeking women
casual encounters
men seeking men
misc romance
missed connections
personals forum new

housing
apts / housing - no fee
apts - broker / fee
apts / housing wanted
rooms / shared
rooms / share wanted
sublets / temporary
sublets wanted
office / commercial
parking / storage
housing directory new

sale / wanted
barter / swap / free
bikes / cycles
cars / trucks
computer / tech stuff
general for sale
furniture new
general wanted
tickets

services
resumes
freelance / 1099
small biz ads

jobs
accounting / finance
admin / office / cust service
architect / engineer / cad
art / media / print / design
business / e-biz / mgmt
education / teaching new
human resources
internet / web engineering
legal / paralegal
marketg / advertis'g / pr
medical / health / biotech
network / telecom
nonprofit sector
retail / hospitality / food
sales / biz dev
skilled trade / craft new
software / QA / DBA
system administration
technical support
tv / film / video / radio
web / info design
writing / editing
et cetera

san francisco
south bay
east bay
north bay
peninsula

boston
new york city
DC
portland_ore
seattle
los angeles
san diego
chicago

sydney
melbourne

The history of Craigslist Personals

INTERNET ARCHIVE <http://www.craigslist.org/> 79,156 snapshots 3 Dec 1999 - 4 May 2005

craigslist **san francisco bay area** [sf](#) [sfbay](#) [chi](#) [aex](#) [aex](#) [aex](#)

post to classifieds
[help](#) [subscriptions](#)

search craigslist

 community

event calendar (2000)
 S M T W T F S
 11 16 17 18 19 20 21
 22 23 24 25 26 27 28
 29 30 31 1 2 3 4
 5 6 7 8 9 10 11

register to vote!
 best-of-craigslist
 craigslist blog
 craigslist T-shirts
 craigslist in the news
 teachers' wishlists
 craigslist foundation
 progressive directory

system status
 NEW! CASH TON! ATTENTION
[about us](#) [contact us](#)
 © 2004 craigslist

community (14100)
[activity, partners](#)
[artists](#) [lost+found](#)
[childcare](#) [musicians](#)
[general](#) [politics](#)
[groups](#) [rideshare](#)
[pets](#) [volunteers](#)
[events](#) [classes](#)

housing (20672)
[apts / housing](#)
[rooms / shared](#)
[sublets / temporary](#)
[housing wanted](#)
[housing swap](#)
[vacation rentals](#)
[parking / storage](#)
[office / commercial](#)
[real estate for sale](#)

for sale (70811)
[barter](#) [babys+kids](#)
[bikes](#) [cars+trucks](#)
[books](#) [clothes+acc](#)
[free](#) [collectibles](#)
[furniture](#) [computer](#)
[general](#) [electronics](#)
[sporting](#) [garage sales](#)
[tickets](#) [household](#)
[wanted](#) [motorcycles](#)
[music instr](#)

services (13499)
[computer](#) [automotive](#)
[creative](#) [household](#)
[erotic](#) [labor/move](#)
[event](#) [skil'd trade](#)
[financial](#) [real estate](#)
[legal](#) [sm biz ads](#)
[lessons](#) [therapeutic](#)

jobs (13026)
[accounting / finance](#)
[admin / office](#)
[art / media / design](#)
[biotech / science](#)
[business / mgmt](#)
[customer service](#)
[education / teaching](#)
[engineering / arch](#)
[human resources](#)
[internet engineering](#)
[legal / government](#)
[marketing / pr / adv](#)
[medical / healthcare](#)
[nonprofit sector](#)
[retail / food / hosp](#)
[sales / biz dev](#)
[skilled trade / craft](#)
[software / qa / dba](#)
[systems / networks](#)
[technical support](#)
[tv / film / video](#)
[web / info design](#)
[writing / editing](#)
[\[ETC\] \[sort time\]](#)

gigs (877)
[computer](#) [event](#)
[creative](#) [labor](#)
[crew](#) [writing](#)
[domestic](#) [talent](#)

resumes (8506)

atlanta [atlanta](#)
[austin](#) [austin](#)
[baltimore](#) [baltimore](#)
[boston](#) [boston](#)
[chicago](#) [chicago](#)
[cleveland](#) [cleveland](#)
[denver](#) [denver](#)
[detroit](#) [detroit](#)
[houston](#) [houston](#)
[los angeles](#) [los angeles](#)
[miami](#) [miami](#)
[minneapolis](#) [minneapolis](#)
[new york](#) [new york](#)
[oklahoma](#) [oklahoma](#)
[philadelphia](#) [philadelphia](#)
[portland](#) [portland](#)
[san diego](#) [san diego](#)
[seattle](#) [seattle](#)
[st. louis](#) [st. louis](#)
[tampa bay](#) [tampa bay](#)
[washington](#) [washington](#)
[wheat ridge](#) [wheat ridge](#)
[winnipeg](#) [winnipeg](#)
[worcester](#) [worcester](#)
[uk:](#) [uk:](#)
[canada:](#) [canada:](#)
[montreal](#) [montreal](#)
[toronto](#) [toronto](#)
[vancouver](#) [vancouver](#)

ARCHIVE
79,156 captures
2 Dec 1998 - 4 May 2025

SEP DEC
NOV
2003 2004 2005

craigslist

post to classifieds
help subscriptions

search craigslist

community >

event calendar (934)

S	M	T	W	T	F	S
1	2	3	4	5	6	
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27

best-of-craigslist
::: craig blog :::
craigslist T-shirts
craigslist in the news
teachers' wishlists
craigslist foundation
progressive directory

system status

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san francisco bay area

city | bkr | sfv | san | oha

community (11309)

- activity partners
- artists lost-found
- childcare musicians
- general politics
- groups rideshare
- pets volunteers
- events classes

personals (48552)

- strictly platonic
- women seek women
- women seeking men
- men seeking women
- men seeking men
- miss romance
- casual encounters
- missed connections
- rants and raves

discussion forums

- aria
- auto
- autos
- bikes
- books
- clothes+acc
- free
- furniture
- general
- sporting
- tickets
- wanted
- asia
- auto
- autos
- bikes
- books
- clothes+acc
- free
- furniture
- general
- sporting
- tickets
- wanted
- asia
- auto
- autos
- bikes
- books
- clothes+acc
- free
- furniture
- general
- sporting
- tickets
- wanted

housing (5514)

apts / housing
rooms / shared
sublets / temporary
housing wanted
housing swap
vacation rentals
parking / storage
office / commercial
real estate for sale

for sale (70477)

- barter
- bikes
- books
- free
- furniture
- general
- sporting
- tickets
- wanted
- barter
- bikes
- books
- collectibles
- computer
- electronics
- garage sales
- household
- motorcycles
- music instr.

services (1556)

- computer
- creative
- erotic
- event
- financial
- legal
- lessons
- automotive
- household
- labor/movg
- skill d trade
- real estate
- sm biz ads
- therapeutic

jobs (12672)

accounting / finance
admin / office
art / media / design
biotech / science
business / mgmt
customer service
education / teaching
engineering / arch
human resources
internet engineering
legal / government
marketing / pr / ad
medical / healthcare
nonprofit sector
retail / food / house
sales / biz dev
skilled trade / craft
software / qa / dba
systems / networks
technical support
tv / film / video
web / info design
writing / editing
(ETC) [part time]

cities (1153)

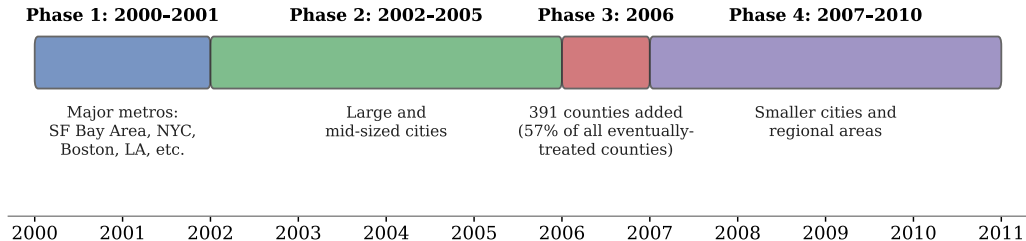
- computer
- creative
- crow
- domestic
- labor
- writing
- talent

resumes (5066)

afanle
albuquerque
anchorage
austin
batonroue
boise
boston
buffalo
charlotte
chicago
cincinnati
cleveland
dallas/fort worth
denver
detroit
fresno
hartford
houston
indianapolis
kansas city
las vegas
los angeles
memphis
miami
minneapolis
new hampshire
new jersey
new york
oklahoma city
omaha
philadelphia
phoenix
portland
providence
raleigh
sacramento
salt lake
san diego
san jose
seattle
st louis
sf bay
spring field
wash DC
canada:
montreal
toronto
vancouver
UK:
london
manchester
edinburgh
IE:
dublin
AU:
melbourne
sydney
32

Craigslist rolled out in geographic phases

Rollout of Craigslist Personals Across U.S. Counties, 2000-2010



Checklist step 1: what will our average causal effect be?

- We are focused on marriage markets, so we target the **unweighted** (or simply weighted) average of county-level treatment effects.
 - Without population weights, you average over the units of observation, here counties, so you estimate the effect for the **average county**.
 - With population weights, large counties get larger and small counties get smaller, so you estimate the effect for the **average woman**.
 - Both are interesting, but they differ, and need not line up or even share a sign under heterogeneous treatment effects.

Checklist step 1: sample window and counterfactual

- We stop the sample in **2007**, because after that we believe the question is probably not answerable.
- We also want $Y(0)$ to be as close as possible to “a world without online dating, but still reasonably popular.”
- **Craigslist** is our solution, and it has the geographic rollout we just saw.

Checklist step 2: make a table of treated units

- First we simply count by cohort.
- With staggered adoption, we need to know how many units are treated and how many are not, because we will have to make decisions later.
- Here is what we initially found, and note it has changed since: a set of counties were treated from 2000 to 2010.
- We don't want to use 2008–2010, because of the huge demographic transition, too many confounders for credible diff-in-diff, and our target-parameter choice.
- So we use the 2008–2010 counties as our primary comparison group, calling them the “**eventually treated.**”

Counts and shares by cohort

Table: Craigslist Personals Expansion and Sample Shares

Craigslist Timing Group	Counties	Share of Treated Counties
2000 cohort	9	0.02
2001 cohort	5	0.01
2002 cohort	12	0.02
2003 cohort	36	0.06
2004 cohort	58	0.10
2005 cohort	69	0.12
2006 cohort	341	0.57
2007 cohort	65	0.11
Total treated counties	595	1.00
Late adopters (2008-2010)	282	–
Total	877	–

This table shows the number of counties that received a Craigslist Personals section in each year as well as the share of treated counties (2000-2007) that each timing cohort makes up.

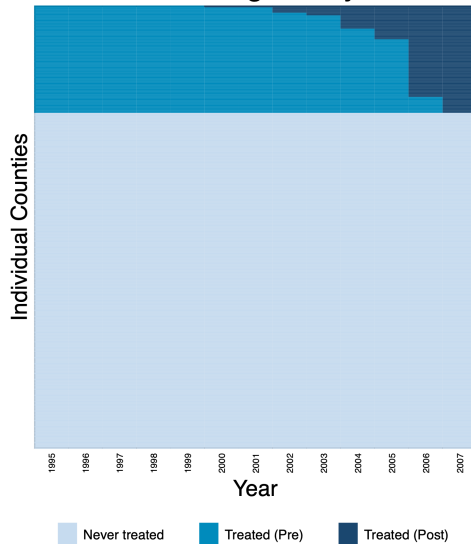
Why group shares and counts

- Why we report cohort shares is a subtle point that is not well known.
- Whenever you average treatment effects across cohorts, you take a *weighted* average based on **cohort size**.
- In an event study, the year-of-treatment ($t = 0$) effect averages over **everyone**, since every cohort has a $t = 0$.
- The 2006 cohort was a huge **57%** of the treated group, so any aggregation including them is heavily swayed by them.
- But the $t = 2$ effect drops the 2006 cohort, because the sample ends in 2007 and they have no $t + 2$ in the data.
- Knowing the group shares tells you that any aggregation is a weighted average using those shares.

Checklist step 3: plot the treatment rollout

- Now create a visualization of the table we just made, plotting the rollout.
- I recommend `PanelView` by Yiqing Xu.
- It plots what you're looking for, and it also flags “missing” cells, so if counties suddenly vanish it reveals it.
- This helps you identify subtle issues ahead of time that are not easily found.
- Remember: you should have $N \times T$ units, with no surprises.

Rollout of Craigslist by Cohort



Checklist step 4: pick your comparison group

- You are usually choosing between the not-yet-treated, the never-treated, and the future-treated.
- Goodman-Bacon (2021) showed that the OLS twoway fixed effects estimator decomposes into a variance-weighted average of every 2×2 in the data, including ones that use the **already-treated** as controls.
- You avoid those by manually computing 2×2 s per cohort with a robust DiD estimator, and we focus on **CS**.
- But you still have to decide who your control group is, and the *only* criterion is: who do I believe parallel trends will hold for?

Checklist step 4: pick your comparison group

- The never-treated and not-yet-treated are more interchangeable the more quasi-random assignment was, since then unobserved selection isn't driving assignment.
- Insofar as you are unsure, I think the **not-yet-treated** is probably better.
- So I actually wrote **Craig Newmark**, who started Craigslist, and asked him.

2:52

5G



New message

craig newmark

decision at all. I don't want to ask you to share things you're not comfortable sharing, and understand if you can't, but I thought I'd ask. Hope you are well.

Scott

1/17/18, 12:43 PM ✓

Scott, thanks! but I haven't been a craigslist spokesman nor in management since 2000, have had no part in decision making. However, you're asking a question involving sophisticated marketing, economic analysis, and business development, and as a matter of public record, craigslist doesn't do that. Thanks!



1/17/18, 12:47 PM

Thanks Craig. That's a



Start a message



Checklist step 4: pick your comparison group

- Nobody knows the selection rules behind treatment assignment if even Craigslist won't tell anyone.
- So I used my best judgment and went with the **2008–2010** group as my comparison.
- My reasoning was simple: at least they got treated eventually.

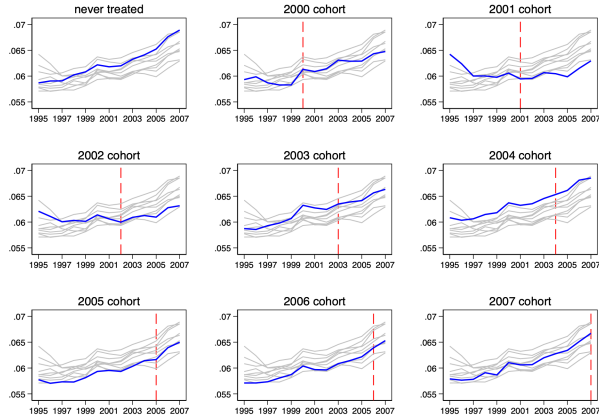
I'm going to skip ahead for the sake of drama.

Checklist step 7: plot outcomes across cohorts

- Make a **beautiful figure** showing the mean outcomes by cohort.
- Remember, we don't care about selection on levels, only selection on **trends**, and we do see signs of it.
- The earlier cohorts have **flatter trends** than the rest.
- Nevertheless, that is what we are doing.

Checklist step 7: plot outcomes across cohorts

Average Births per 1,000 females by Cohort
15-44 year olds



Checklist step 8: estimator selection and assumptions

- **Estimator choice:** choose an estimator based on comfort and assumptions.
- **Robust method:** use robust methods like CS, SA, dCDH, or BJS if agnostic to heterogeneous effects.
- **Event study:** estimate simple averages as well as event studies.
 - Here I used the default syntax in Stata for event studies.
- **Cluster standard errors:** report with clustering at the aggregate level of treatment.

The Callaway & Sant'Anna estimator (the IPW version)

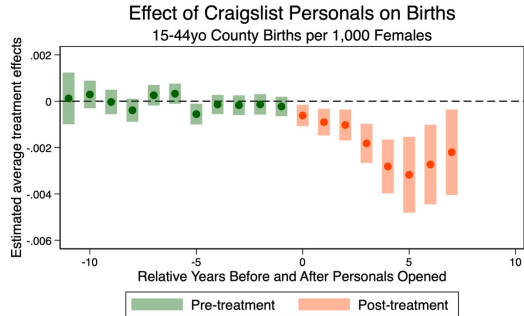
$$ATT(g, t) = \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{\hat{p}(X) C}{1 - \hat{p}(X)}}{\mathbb{E} \left[\frac{\hat{p}(X) C}{1 - \hat{p}(X)} \right]} \right) (Y_t - Y_{g-1}) \right]$$

- There is an IPW version, an outcome-regression version, and a doubly-robust version, and Sant'Anna recommends **doubly robust**.
- CS uses the never-treated or the not-yet-treated as controls, **never the already-treated**.

My choices

- I will use **CS** because it is simple (four averages and three subtractions) and popular.
- I will not use covariates.
- I will make an event study.

Short-gaps, no controls



Note: Uses not-yet-treated including the 2008-2010 but not the never treated counties (i.e., model specification 2 from table estimates). Circles are average ATT(g,t) per relative time period across all groups and bands are 95% uniform confidence intervals. Mean birth rate variable was approximately 0.06 in 2000.

Figure 9: Event study plots estimated using Callaway and Sant'Anna. Comparison group is the not-yet-treated cohorts, including the 2008-2010 cohorts who were treated after the end of our panel period. Data is from 1995 to 2007 and based on county level data.

Doubt ensued

- The event study plots persuaded us that unconditional parallel trends was reasonable.
- But then I learned about a “software default” in both R and Stata.
- If you do not specify `long2` in `csdid`, or `base_period="universal"` in R, the default is short gaps.
- I had always taught long differences, and it never occurred to me that this wasn't doing that (double-check the 2×2).
- But my event study plots looked so good, and I already had a narrative about “permanent dating”!

Trends or no trends?!

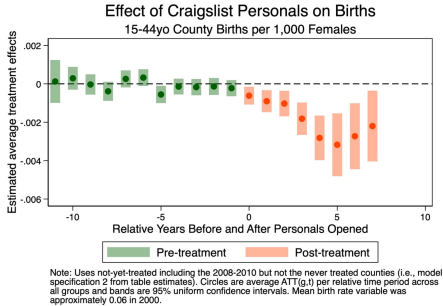
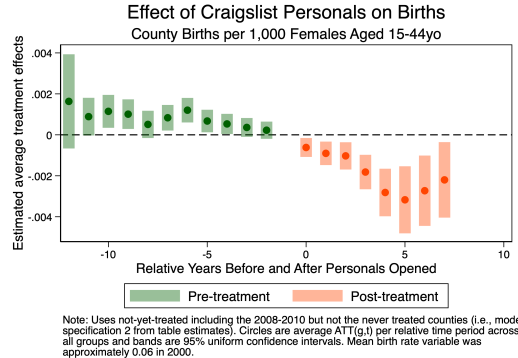


Figure 9: Event study plots estimated using Callaway and Sant'Anna. Comparison group is the not-yet-treated cohorts, including the 2008-2010 cohorts who were treated after the end of our panel period. Data is from 1995 to 2007 and based on county level data.

Short gap



Long difference

Scrutinizing that event study

- I was so excited when I saw those flat pre-trends, until I learned that the default syntax in Stata's `csdid` calculated short gaps.
- We had a whole story that online dating caused “permanent dating,” until I realized that, lol.
- But that made us start wondering whether unconditional parallel trends was plausible.
- We started to piece together what our different units were, that is, “counties” who were treated versus “counties” who weren't.
- Why would some counties get a Craigslist but not others?

Who got a Craigslist and who didn't?

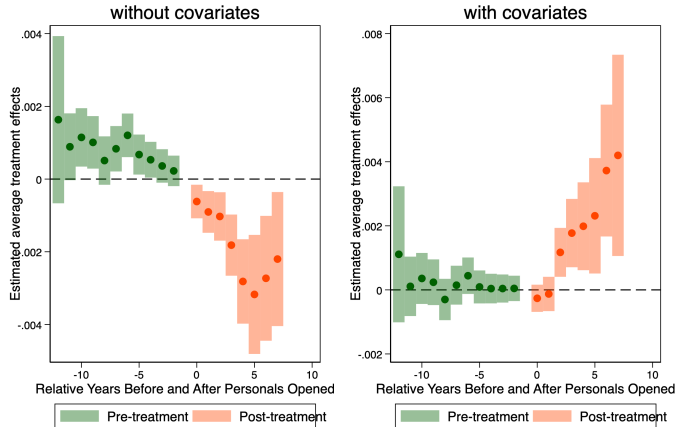
- Craigslist is organized by city, like `houston.craigslist.org` for Houston, Texas.
- I grew up in Brookhaven, Mississippi, a town of 11,000, and `brookhaven.craigslist.org` is not there.
- Craigslist targeted cities, but most US counties are rural.

Is unconditional parallel trends plausible?

- My coauthors are demographers with a 20-year research agenda in maternal health and children.
- We are all labor economists, familiar with “sorting” into rural versus cities and how population characteristics differ:
 - Cities have more college-educated women, delayed childbearing, lower marriage rates, different racial composition, and more access to female healthcare.
 - Rural towns have less-educated women, earlier age at first birth, higher marriage rates, more homogeneity, and worse healthcare access.
- If Craigslist counties are more urban, then treatment and control are imbalanced on covariates that cause trends in Y^0 .
- So we looked at mean female population and urban measures by timing cohort to see how bad the covariate imbalance was.

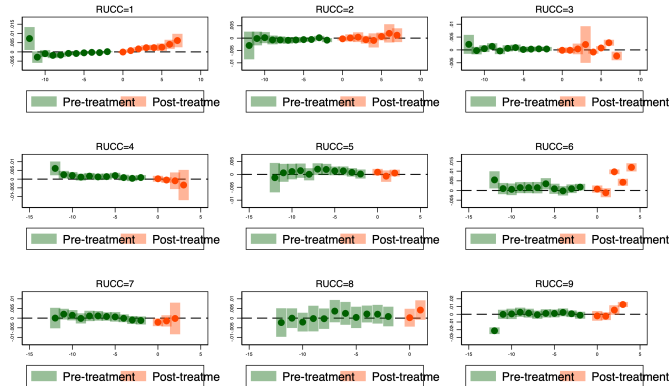
Long differences without and with RUCC control (!)

Event Study on br1544



Running it within each RUCC code

Estimated Effect of Craigslist Personals on Birth Rates by RUCC codes



Smaller RUCC codes are more urbanized counties. Models estimated separately with CS and not-yet-treat

Checklist step 3: validity of unconditional parallel trends

3. Validity of the unconditional parallel trends assumption

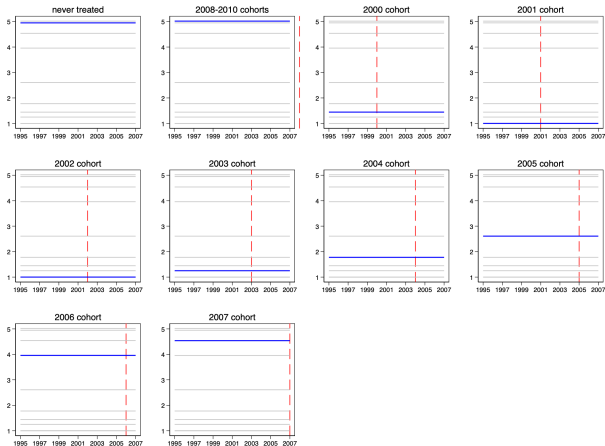
- **Covariate selection:** choose covariates predictive of the missing potential outcome based on common sense, literature, and expertise.
- **Split-sample validation:** use untreated outcomes with methods like Lasso to identify covariates if unsure.
- **Balance check:** check covariate balance using normalized differences.
- **Adjust imbalanced covariates:** adjust if covariates are predictive of $[Y^0 \mid D=1, \text{Post}]$.

Rural vs. urban counties

- We have a 9-digit ordinal score measuring how urbanized a county is, called **RUCC**.
- Counties that never got a Craigslist were **more rural**.
- We had originally wanted to use the 2008–2010 and other not-yet-treated counties as comparisons.
- I'll **plot** rather than report the normalized difference, for illustrative purposes.

Plotting covariate imbalance

Average 2003 RUCC Code by Cohort



Comment

- Urban counties differ in race, age, sex ratios, education shares, and income, all **highly predictive** of untreated birth-rate trends.
- And even the 2008–2010 counties, despite our hope, are very rural, just as rural, even.
- So we will run it **both ways**: without controls, and then with a single covariate (RUCC).

Checklist step 4: check overlap

4. Check overlap

- **Overlap check:** ensure sufficient overlap in propensity scores.
- **Adjustment for lack of overlap:** if lacking, consider a regression adjustment like BJS.
- We have overlap since the RUCC code has only 9 values, so I'll skip it here, but you'll want to check using a **propensity score** if you have continuous or several covariates.

Watch for perfect separation. With a propensity-score model, a covariate that perfectly predicts treatment forces $\hat{p}(X) \rightarrow 0$ or 1, weights blow up, and the estimator is undefined. Few discrete covariate values (like RUCC) keep you safe; rich or continuous covariates do not.

Checklist step 6: sensitivity analysis for PT violations

6. Sensitivity analysis for parallel-trends violations

- **Consider sensitivity analysis:** use tools like `honestdid` for PT violations.

Checklist step 7: throw in the towel

7. Throw in the towel

- You may have to concede that, for any number of reasons, parallel trends with or without covariates is implausible, it happens.
- Consider some alternatives:
 - **Panel method:** if imbalance or PT fails, consider matrix completion or nuclear-norm regularization.
 - **Cross-section:** cross-sectional unconfoundedness may also be appropriate.
- Don't force it, diff-in-diff doesn't make estimates causal; **parallel trends makes diff-in-diff causal.**

LESSON

Concluding Remarks

Concluding remarks on DiD

- You're probably going to write a paper using DiD at least once in your life, probably more.
- Even if you don't, you'll read many DiD papers, referee them, or advise students using them.
- It's in your interest to make the fixed-cost investment in the new econometrics of DiD, because the old methods are mostly harmful.
- Good news: we are at the conclusion of this wave of papers, software is now widely available, and solutions share common features, static and dynamic presentations aren't all that different.
- Lots of surveys exist (Roth et al. 2023; our new JEL), and you'll have the slides and recordings.

Concluding remarks: covariates

- Stronger assumptions are needed to include covariates, and the bias can be large.
- You need the **correct** covariates, I suggested sample splitting for selection and Lasso to identify covariates that affect **untreated potential outcomes**.
- Don't control for covariates that could be affected by the outcome.

Concluding remarks: differential timing

- The main problem in differential timing is heterogeneity and the use of already-treated units as controls.
- With differential timing and no a priori theory, you *should* avoid the canonical TWFE specification.
- Under heterogeneity, canonical TWFE is biased, signs can flip, weights can be negative.
- Robust DiD methods place no restrictions on treatment-effect heterogeneity, but they can't solve everything.
- When you lose parallel trends, you're probably heading toward needing a new design.

Concluding remarks: design discipline

- Devote time to **designing** your diff-in-diff, it keeps you honest and surfaces problems before you've peeked at the outcomes.
- It's much harder, psychologically, to pull back once you're invested in a story, so don't go to the outcomes until you've pre-registered your study to some degree, even if not formally.

Diff-in-diff doesn't make estimates causal.
Parallel trends makes diff-in-diff causal.
